

PART A.

1. Faraday's law of induction states that "The induced emf, V_{emf} (in volts) in any closed circuit is equal to the time rate of change of the magnetic flux linkage by the circuit).
2. Lenz's law states that "The direction of current flow in the circuit is such that the induced magnetic field produced by the induced current will oppose the original magnetic field".
3. Equation of transformer emf

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\oint_C (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} = - \int_C \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

4. motional electric field is the induced emf is an electric field that appears in a conductor's rest frame when the conductor moves through a magnetic field.

5. motional emf is the induced emf due to the motion of the electric conductor in the presence of the magnetic field.

6. emf produced by moving loop in time varying field is the rate of change of the magnetic flux through loop:

7. time harmonic field $\div$ is a field that varies periodically or sinusoidally with time at a specific angular frequency.

Point form	Integral form	Significance
$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \oint \vec{B} \cdot d\vec{S}$	Faraday's law
$\nabla \times \vec{H} = \vec{J}$	$\oint \vec{H} \cdot d\vec{l} = I = \int \vec{J} \cdot d\vec{S}$	Ampere's Circuital law
$\nabla \cdot \vec{D} = 0$	$\oint \vec{D} \cdot d\vec{S} = 0$	Gauss's law
$\nabla \cdot \vec{B} = 0$	$\oint \vec{B} \cdot d\vec{S} = 0$	No isolated magnetic charges.

1. Significance of Displacement Current density.

Displacement current density is significant because it is source of magnetic field. The following are significance of displacement current density.

(i) Continuity of the magnetic field.

Displacement current is responsible for the continuity of the magnetic field between the plates of a charging capacitor.

(ii) Electromagnetic waves.

Displacement current is a key part of the derivation of electromagnetic waves, which are the basis for the theory of light.

(iii) Conservation of charge.

Displacement current makes Ampere's law consistent with the principle of conservation of charge.

(iv) Maxwell's equations.

Displacement current is a crucial addition to Maxwell's equations, which unite electricity, magnetism

and optics into a single theory.

10. Ways of EMF generation.

(i) Electromagnetic induction.

This process creates EMF by moving a magnetic field around an electric conductor or moving an electric conductor through a magnetic field.

(ii) Motional EMF

This ~~is~~ ^{occurs} when EMF is induced due to the motion of an electric conductor in a magnetic field.

(iii) Self-induced EMF

This occurs when EMF is induced in a coil due to a change in flux produced by linking it with its own turns.

(iv) Changing the magnetic field

This can be done by changing the current flowing in the electromagnet that produces the magnetic field.

(iv) Changing the angle.

This can be done by changing the circuit's loop in the magnetic field.

(i), Practical eg. generators due to rotating a coil in a magnetic field.

$$\epsilon = - \frac{d\Phi_B}{dt} = - \frac{d(B \cdot A \cdot \cos \theta)}{dt}$$

(ii) Practical eg. Electromagnetic braking, Railguns and Energy Harvesting, electric generator.

$$\epsilon = B \cdot L \cdot v \cdot \sin \theta$$

(iii) Practical eg. Relay and motor circuits, inductive loads, Energy storage in inductors

$$\epsilon_{\text{self}} = - L \frac{dI}{dt}$$

(iv) Practical eg. voltage step-up/step-down and inductive coupling.

$$\epsilon = - \frac{d\Phi_B}{dt} \quad \Phi_B = B \cdot A \cdot \cos \theta$$

(1) Practical eg, power plants, and
Hand-crank Generators.

e_m

$$\Phi_B = B \cdot A \cdot \cos \theta$$
$$e = - \frac{d\Phi_B}{dt}$$

but $\Phi_B = B \cdot A \cdot \cos \theta$

$$e = - \frac{d(B \cdot A \cdot \cos \theta)}{dt}$$

Uniform plane wave.

* Waves is the means of transporting energy or information from source to destination.
example electromagnetic waves which consist of electric and magnetic field.

* Properties of electromagnetic waves

(i) They assume properties of waves while travelling.

(ii) They travel with high velocity.

(iii) They radiate outwards from source in all directions.

$$\nabla^2 \vec{E} = \mu \sigma \frac{\partial \vec{E}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{H} = \mu \sigma \frac{\partial \vec{H}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

For free space $\sigma = 0, \epsilon = \epsilon_0, \mu = \mu_0$

$$\begin{cases} \nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \\ \nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2} \end{cases} \quad \eta = \mu_0 \sqrt{\frac{\mu_r}{\epsilon_r}}$$

Prob 1.10 Given $\epsilon_r = 8, \mu_r = 2$

$$\vec{E} = 0.5 \sin(10^8 t - z) \vec{a}_z \text{ V/m}$$

(i) Loss tangent

(ii) Wave impedance

(iii) Wave velocity

Soln.

$$\beta = \omega \sqrt{\mu \epsilon_{\text{ref}}}$$

$$\eta = \frac{E}{H} \quad E_i + E_r = E_t \quad H = \frac{E}{\eta}$$

$$H_i + H_r = H_t$$

$$\frac{E_i}{\eta_i} + \frac{(-E_r)}{\eta_r} = H_t$$

$$\frac{1}{\eta_1} (E_i - E_r) = H_t \quad \text{--- (1)}$$

$$H_t = \frac{E_t}{\eta_2} = \frac{E_i + E_r}{\eta_2} \quad \text{--- (2)}$$

$$\frac{1}{\eta_1} (E_i - E_r) = \frac{E_i + E_r}{\eta_2}$$

$$\eta_2 E_i - \eta_2 E_r = \eta_1 E_i + \eta_1 E_r$$

$$E_i (\eta_2 - \eta_1) = E_r (\eta_1 + \eta_2)$$

$$\frac{E_r}{E_i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$E_{(2)} = E$$